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BATCH : B4 -B2 ROLL.NO: 31

SUB : SPL PROJECT

1. Title

Spam SMS Detection using Machine Learning

2. Project Objective

The main objective of this project is to develop a machine learning model that can classify SMS messages as **spam** or **ham (not spam)**.
The system aims to:

- Improve message filtering
- Reduce unwanted spam messages
- Provide a user-friendly interface for real-time prediction

3. Introduction

With the rapid growth of mobile communication, spam messages have become a major issue. These messages often contain advertisements, fraud attempts, or malicious links.

Machine Learning helps automate spam detection by analyzing text patterns and learning from historical data. This project uses Natural Language Processing (NLP) techniques along with classification algorithms to detect spam messages efficiently.

4. Project Methodology

Steps followed:

1. Data Collection
2. Data Preprocessing
3. Feature Extraction (TF-IDF)
4. Model Training
5. Model Evaluation
6. Model Comparison
7. Best Model Selection
8. GUI Development

5. Project Implementation

1. Data Preprocessing

- Removed null values
- Removed duplicates
- Converted text to lowercase
- Removed stopwords and punctuation

2. Feature Extraction

- Used **TF-IDF Vectorizer** to convert text into numerical form

3. Model Training

Models used:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)

4. Model Evaluation

Metrics used:

- Accuracy
- Confusion Matrix

5. GUI Development

- Built using **Tkinter**
- Allows user to input SMS and get prediction

7. Results and Discussion

1) Data preprocessing Output

```
[14]
import string

def clean_text(text):

    text = text.lower()

    text = text.translate(str.maketrans('', '', string.punctuation))

    return text

df['clean_message'] = df['message'].apply(clean_text)
```

```
[15]
data = df.dropna()
```

```
[16]
print(df.isnull().sum())

label            0
message          0
clean_message    0
dtype: int64
```

```
[17]
df = df.drop_duplicates()
```

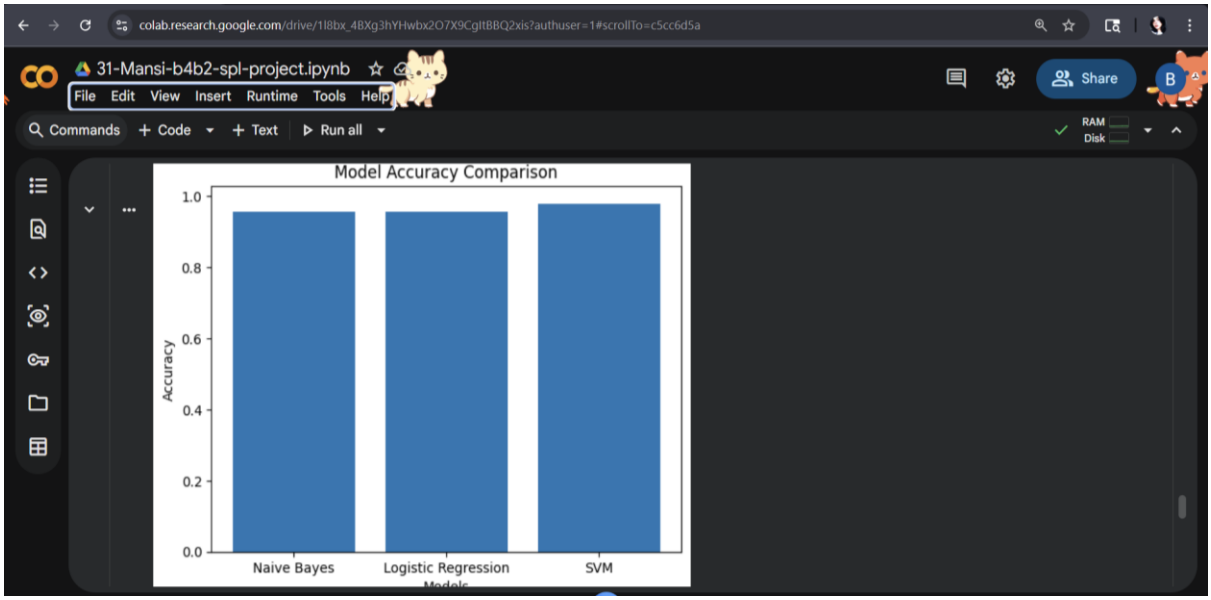
2) Model comparison table

```
[25]
from sklearn.metrics import accuracy_score

print("Naive Bayes Accuracy:", accuracy_score(y_test, pred_nb))

print("Logistic Regression Accuracy:", accuracy_score(y_test, pred_lr))

print("SVM Accuracy:", accuracy_score(y_test, pred_svm))

Naive Bayes Accuracy: 0.9623318385650225
Logistic Regression Accuracy: 0.9632286995515695
SVM Accuracy: 0.9766816143497757
```

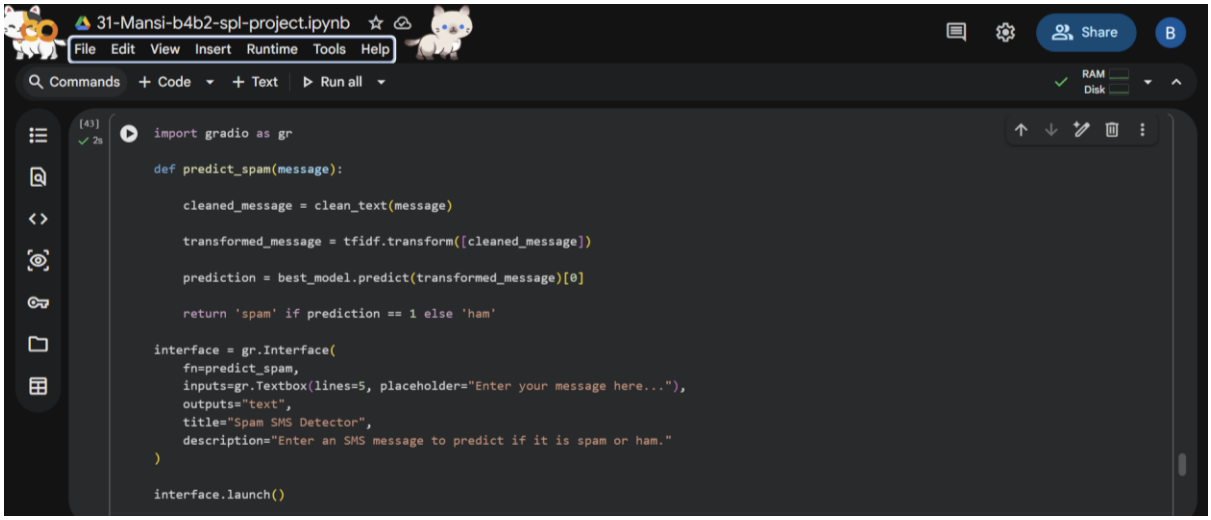
```
[34]
best_model_name = max(results, key=results.get)
best_accuracy = results[best_model_name]

print("\nBest Model is:", best_model_name)
print("Accuracy:", best_accuracy)

if best_model_name == "Naive Bayes":
    best_model = nb
elif best_model_name == "Logistic Regression":
    best_model = lr
elif best_model_name == "SVM":
    best_model = svm

Model Accuracies:
Naive Bayes : 0.9555125725338491
Logistic Regression : 0.9555125725338491
SVM : 0.9796905222437138

Best Model is: SVM
Accuracy: 0.9796905222437138
```

3) Confusion matrix

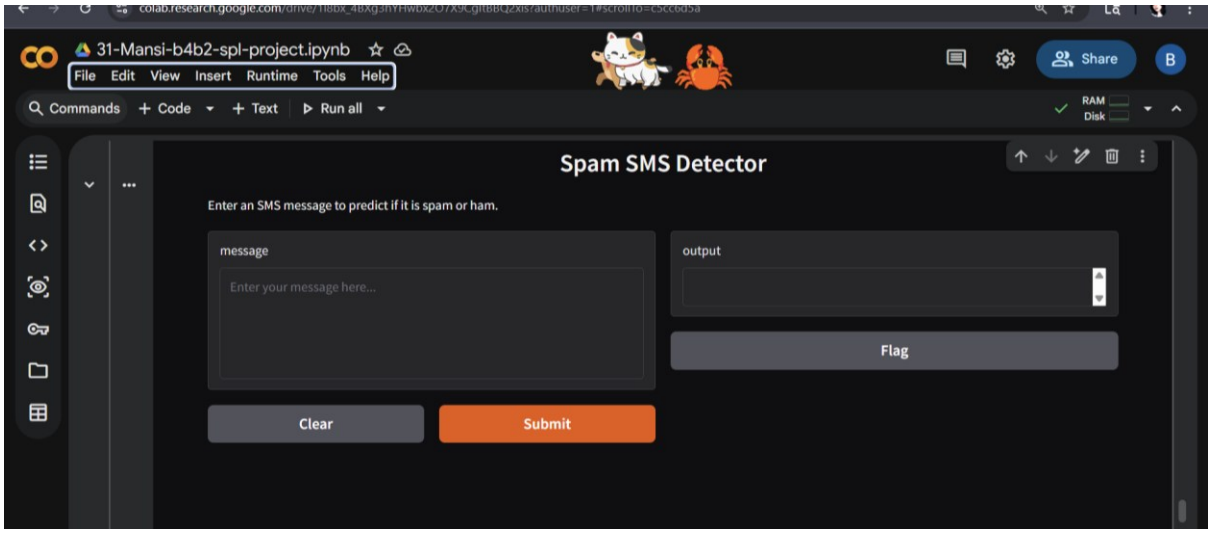
4) accuracy comparison



5) GUI interface



6) GUI implementation



8.Conclusion

This project successfully demonstrates how machine learning can be used to detect spam SMS messages. Among the tested models, the best-performing model achieved the highest accuracy and was integrated into a GUI for real-time predictions.

Future improvements can include:

- Deep learning models
- Better dataset
- Web-based deployment

9. GitHub Link

https://github.com/24lacchhodem-ship-it/spam-sms-detection.git